Numerical Solution of a Fredholm Integral Equation of the Second Kind

Mathematical Analysis

1. Problem formulation

Consider the Fredholm integral equation of the second kind:

$$x(t) = \lambda \int_a^b K(t,s)x(s)\,ds + y(t)$$

For this problem:

$$x(t) = 0.6 \int_{0.3}^{1.2} t^2 s^2 x(s)\,ds + \sin(t)/t$$

Parameter	Value
Lower integration limit, a	0.3
Upper integration limit, b	1.2
Kernel, K(t,s)	t^2s^2
Known function, y(t)	$\sin(t)/t$
Parameter, λ	0.6
Simpson subinterval pairs, n	9
Total subintervals, N = 2n	18

Objective: investigate the contraction mapping principle, derive the analytical solution, determine the required number of successive approximations, and implement the numerical method in Python.

2. Part 1 — Verification of the contraction mapping principle

Define A by $(Ax)(t) = \lambda \int_a^b K(t,s)x(s)\,ds + y(t)$. It acts on $(C[a,b], \rho)$, where $\rho(x,z) = \max_{t\in[a,b]} |x(t) - z(t)|$.

Step 1. Establish the contraction inequality

$$|(Ax)(t) - (Az)(t)| \le |\lambda| t^2 \rho(x,z) \int_a^b s^2\,ds$$

$$\rho(Ax,Az) \le |\lambda| \max_{t\in[a,b]}(t^2) \left(\int_a^b s^2\,ds\right) \rho(x,z)$$

Step 2. Calculate the contraction constant

$$\int_{0.3}^{1.2} s^2\,ds = [(s^3)/3]_{0.3}^{1.2} = (1.728 - 0.027)/3 = 0.567$$

$$\max_{t\in[0.3,1.2]} t^2 = 1.2^2 = 1.44$$

$$\theta = 0.6 \times 1.44 \times 0.567 = 0.489888$$

Conclusion: Since $0 \le \theta = 0.489888 < 1$, A is a contraction. The space C[0.3,1.2] with the uniform metric is complete, and A maps this space into itself. By the Banach fixed-point theorem, the equation has a unique continuous solution.

3. Part 2 — Derivation of the analytical solution

$$I = \int_{0.3}^{1.2} s^2 x(s)\, ds$$

$$x(t) = 0.6t^2 I + \sin(t)/t$$

Step 1. Substitute x(t) into I

$$I = 0.6I \int_{0.3}^{1.2} s^4\, ds + \int_{0.3}^{1.2} s \sin(s)\, ds$$

Step 2. Evaluate the first integral

$$\int_{0.3}^{1.2} s^4\, ds = [(s^5)/5]_{0.3}^{1.2} = (2.48832 - 0.00243)/5 = 0.497178$$

$$0.6 \times 0.497178 = 0.2983068$$

Step 3. Evaluate the second integral by integration by parts

$$\int s \sin(s)\, ds = -s \cos(s) + \sin(s)$$

$$J = [-s \cos(s) + \sin(s)]_{0.3}^{1.2} = 0.4882905207$$

Step 4. Solve for I

$$I = 0.2983068I + 0.4882905207$$

$$I = 0.4882905207 / (1 - 0.2983068) \approx 0.6958746653$$

$$0.6I \approx 0.4175247992$$

Analytical solution:

$$x(t) = 0.4175247992t^2 + \sin(t)/t$$

4. Part 3 — Determination of the number of successive approximations

$$x_{k+1}(t) = 0.6t^2 \int_{0.3}^{1.2} s^2 x_k(s)\, ds + \sin(t)/t$$

$$x_0(t) = \sin(t)/t$$

Step 1. Calculate the first approximation

$$x_1(t) = 0.6t^2 \int_{0.3}^{1.2} s \sin(s)\, ds + \sin(t)/t$$

$$x_1(t) = 0.2929743124t^2 + \sin(t)/t$$

Step 2. Calculate the distance between the first two approximations

$$\rho(x_0, x_1) = 0.2929743124 \times 1.44 = 0.4218830099$$

Step 3. Apply the a priori error estimate

$$\rho(x^*, xL) \leq [\theta^L/(1-\theta)]\, \rho(x_0, x_1)$$

$$L > \ln(0.01(1 - 0.489888)/0.4218830099) / \ln(0.489888) \approx 6.1875015$$

The minimum integer number of iterations is Lmin = 7. For an additional accuracy margin, choose L = 8.

$$\rho(x^*, x_8) \leq 0.0027435 < 0.01$$

5. Part 4 — Python implementation using composite Simpson's rule

The integration interval is divided into N = 2n = 18 equal subintervals. The step size is h = (1.2 − 0.3)/18 = 0.05.

$$\int_a^b f(s)\,ds \approx (h/3)[f(s_0)+f(sN)+4\Sigma\text{odd } f(si)+2\Sigma\text{even } f(si)]$$

```
import numpy as np

# 1. Problem parameters
a = 0.3
b = 1.2
lam = 0.6
n = 9
L = 8

# 2. Simpson's rule discretization
N = 2 * n
h = (b - a) / N
t = np.linspace(a, b, N + 1)

# 3. Known function and initial approximation
y = np.sin(t) / t
x = y.copy()

# 4. Simpson's rule weights
w = np.ones(N + 1)
w[1:N:2] = 4
w[2:N:2] = 2

# 5. Successive approximation method
for k in range(L):
    x_new = np.zeros_like(x)

    for i in range(N + 1):
        kernel = t[i]**2 * t**2
        integral = (h / 3) * np.sum(w * kernel * x)
        x_new[i] = lam * integral + y[i]

    x = x_new

# 6. Analytical solution
J = (-b * np.cos(b) + np.sin(b)) \
    - (-a * np.cos(a) + np.sin(a))

A = (b**5 - a**5) / 5
I = J / (1 - lam * A)

exact = lam * I * t**2 + y

# 7. Maximum absolute error
R = np.max(np.abs(exact - x))

# 8. Display results
theta = lam * b**2 * (b**3 - a**3) / 3

print("Number of iterations:", L)
print("Step size:", h)
print("Contraction constant:", theta)
print("Exact solution coefficient:", lam * I)
print("Maximum absolute error R:", R)
```

Expected numerical output

```
Number of iterations: 8
```

```
Step size: 0.05
Contraction constant: 0.489888
Exact solution coefficient: 0.4175247992
Maximum absolute error R: 3.74037e-05
```

The maximum absolute error is approximately $R = 3.74037 \times 10^{-5}$, which is smaller than the prescribed tolerance of 0.01.

6. Final conclusion

The Fredholm integral equation was investigated using the contraction mapping principle and the method of successive approximations. The contraction constant was found to be $\theta = 0.489888 < 1$, proving the existence and uniqueness of the continuous solution.

$$x(t) = 0.4175247992t^2 + \sin(t)/t$$

The a priori error estimate showed that at least seven iterations are required to achieve the prescribed accuracy. Eight iterations were performed using Python and the composite Simpson's 1/3 rule. The resulting maximum absolute error was approximately 3.74037×10^{-5}, confirming that the numerical approximation satisfies the required accuracy.